

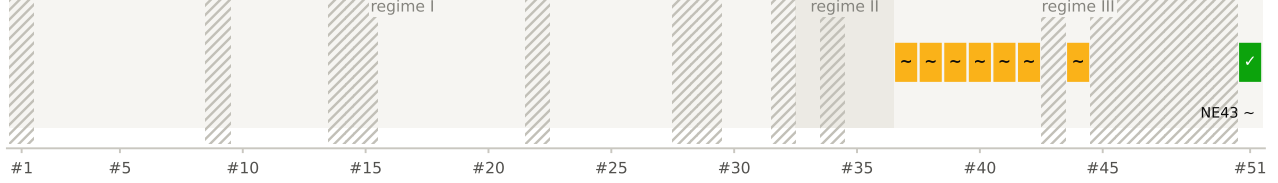
H90: Collective entropy production beyond the parts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

Bottom line. The swarm's only measurable collective arrow of time is agents talking in the minute after the agents who named them talked; it is coupling between named pairs, not a new collective variable, and only a 45-day period can resolve it.

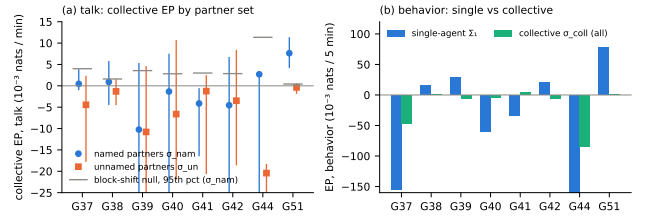
level: hypothesis · pipeline checks 86% · updated 2026-10-04



Goal periods tested (1 supported, 8 mixed). ✓ supported ✗ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

I measured whether the swarm's joint dynamics are more irreversible than its agents' own dynamics. The estimator is the Aguilera-Ito-Kolchinsky max-ent bound on DQ8-trimmed windows, in three channels (5-min behavior states, 1-min talk, 1-min activity), on eight regime-III periods. Synthetic validation came first. In G51 (45 days, 21–32 agents) the talk channel carries a collective arrow of time: $\sigma_{\text{nam}} = 7.6 [4.3, 11.2] \times 10^{-3}$ nats/min on named partners, none on unnamed partners (block-shift $p = 0.01$). Single-agent talk is reversible, so this arrow is fully collective. Behavior is irreversible per agent (0.078 nats per 5 min) with a collective share of 0.003. The seven smaller periods are underpowered. Pairwise directions do not follow who names whom. Holdout (#51 tail): not run.



Observables. (a) Talk collective EP per period with the named-partner (blue) and unnamed-partner (orange) fields, day-bootstrap 95% CIs, and the block-shift null's 95th percentile (grey). Only G51 clears the null, on named partners only. (b) Behavior: single-agent bound Σ_1 (blue) and collective term (green). G51's single-agent arrow is clear; its collective term is 2×10^{-4} . G37 and G44 bars are clipped below the floor.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Two uses, both modest. First, a monitor that wants to detect agents coordinating should look at talk timing between agents that name each other: that is where the swarm's only joint arrow of time sits, at about 0.017 nats per agent-hour in #51. Second, a method warning: a block-shift or cross-day null does not separate coupling from a common drive with agent-specific lags; split partners by addressing instead. The detector needs about 40 days and 25 agents. On a 5-day, 15-agent week it has no power (power ≤ 0.05 at the #51 effect size). There is no egregore here: removing the naming partition removes the collective term.

PHYSICAL MODEL

Models 15 (EP inference) and 02 (kinetic spins with asymmetric couplings). Spins: an agent's behavior state (5 soft states) or its talk or activity in a minute. Single observables u_i : one agent's time-antisymmetric patterns. Coupling observables g_i : agent i 's state against a partner field h_i , the mean over agents who named i (named), never named either way (unnamed), or all others. Σ_g : the max-ent lower bound on entropy production for observables g ; θ : its multipliers.

$$\begin{aligned} \Sigma &\geq \Sigma_g = \max_{\theta} [\theta^T \langle g \rangle - \ln \langle e^{-\theta^T g} \rangle] \\ \Sigma_g &\approx 2 \langle g \rangle^T K^{-1} \langle g \rangle \quad (\text{held out by day fold}) \\ g_i(t) &= s_i(t+1) h_i(t) - s_i(t) h_i(t+1) \\ h_i^{\text{nam}} &= \sum_j w_{ji} s_j / \sum_j w_{ji} \quad (w_{ji} : j \text{ names } i) \\ \sigma_{\text{coll}} &= \Sigma_{u \cup g} - \Sigma_u \geq 0, \quad \Delta_{\text{addr}} = \sigma_{\text{nam}} - \sigma_{\text{un}} \end{aligned}$$

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 $\rho_{\text{coll}} < 0.1$ (behavior, activity)	G51 behavior 0.003; activity $\Sigma \approx 10^{-4}$	supported
P2 talk $\sigma_{\text{coll}} > \text{shift null}$	G51 $p = 0.01$; 0/7 elsewhere	G51 only
P3 $\Delta_{\text{addr}} > 0$ (coupling test)	G51 $8.0 [5.1, 12.1] \times 10^{-3}$, $p = 0.01$; G38 $p = 0.08$	supported where powered
P4 rank vs H50 J_1	Spearman -0.29 ($n = 8$)	failed
P5 $\Sigma_1 \approx \sum_i \sigma_i$	gap ≤ 0.033 nats/step	supported
N1 G38 same room $>$ cross room	$+1.6$ ($p.03$) vs -0.7 ($p.66$) $\times 10^{-3}$	supported (weak)
N2 G51 pair θ follows naming	agreement 0.52 (null 0.55)	failed
N3 NE43 nudger off	σ_{coll} in placebo range; $\sigma_{\text{nam}} \times 0.80$	mixed

GOAL PERIODS IN DETAIL

G37	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G38	$\sim$ mixed (replication underpowered; native N1 rooms: sup...)
G39	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G40	$\sim$ mixed (no address contrast; one single-channel be-havi...)
G41	$\sim$ mixed (no address contrast; one single-channel be-havi...)
G42	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G44	$\sim$ mixed (underpowered at the G51 effect size; "failed")
...	...
G51	$\checkmark$ supported (native N2 in this folder: failed)
NE43	$\sim$ mixed (σ_{coll} (all) inside the placebo range; σ_{nam} fa...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

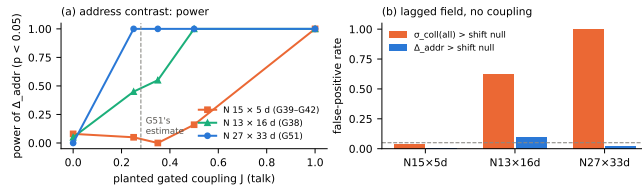
CAVEATS

One period carries the result. The 1-min talk bins put most read-out responses in the same bin as their cause, so the bound misses them and reads low. Naming uses any roster mention, not the leading-@ target. The block-shift null fails against lagged fields, so only the address contrast is a coupling claim. The estimator was fixed after a numerical failure on real behavior data (A2), then re-validated. The power reading of the small periods is post hoc (A3). No read vs unread split of the naming messages.

ROUND 2

Rebuild talk spins on each agent's call clock with the read-out call as origin, which should make 5-day periods testable. Split σ_{nam} by read vs posted-but-unread naming messages. Run `confirm.py` on the #51 tail after registering H90 in the holdout ledger.

- **Essence.** whether the swarm has collective order that no agent carries alone, measured as irreversibility, with the fields removed.
- **H90-R1** Call-clock talk spins.
- **H90-R2** Read vs posted-but-unread partner fields.
- **H90-R3** Leading-@ targets.
- **H90-R4** Hierarchical pooling across 5-day periods (exception (d)): a shrinkage estimate of Δ_{addr} per period, reported next to the per-period values.
- **H90-R5** Run `confirm.py` after registering H90 in the holdout ledger.



Second figure. Synthetic agents at the three real sizes with soft labels. (a) Power of the address contrast against planted gated coupling J ; the dashed line is G51's estimate ($J \approx 0.28$). (b) A common drive with agent-specific lags and no coupling: the collective term beats the block-shift null in 63–100% of large worlds, the address contrast in at most 10%.